BRSM Mini Project Phase-1

Analysis of Advertisement data

Source: https://www.kaggle.com/datasets/tawfikelmetwally/advertising-dataset/data

1. Introduction

1.1 Background

Advertising is a critical component of marketing strategies, enabling businesses to promote products and drive sales. Understanding the effectiveness of different advertising platforms (TV, Radio, Newspaper) is essential for optimizing marketing budgets and maximizing returns. This project analyzes a dataset containing historical advertising spends and corresponding sales data to identify which platforms yield the highest return on investment (ROI) and uncover trends that can inform future advertising strategies.

1.2 Dataset Overview

The dataset consists of 200 observations with the following variables:

- TV: Advertising spend on TV (in thousands of dollars).
- Radio: Advertising spend on Radio (in thousands of dollars).
- Newspaper: Advertising spend on Newspaper (in thousands of dollars).
- Sales: Corresponding sales (in Million dollars).

The dataset is sourced from a marketing campaign conducted by a retail company. The goal is to analyze the impact of advertising on sales and derive actionable insights.

2. Methods

2.1 Data preparation

- The dataset was loaded from the given link (from Kaggle).
- An unwanted column “Unnamed: 0” was removed from the dataset.
- Descriptive statistics were computed to summarize the data.

2.2 Descriptive Statistics

Descriptive statistics were calculated to summarize the central tendency and dispersion of the variables. Metrics included:

- Mean: To understand average spending/sales.
- Median: To assess central value robustness against outliers.
- Standard Deviation: To quantify variability in ad spends and sales.
- Range (Min/Max): To identify extreme values or potential outliers.

Why These Metrics?

The mean and median provide a baseline understanding of typical spending and sales. Standard deviation and range highlight variability, which is crucial for identifying outliers (e.g., unusually high Newspaper spends).

2.3 Visualizations

Histograms & Kernel Density Estimation (KDE):

- What: We plotted histograms overlaid with KDE curves for each variable.
- Why: Histograms reveal the frequency distribution, while KDE offers a smooth estimate of the data distribution. This dual view helps us detect skewness, multimodality, and the overall shape of the data.

Box Plots:

- What: We used box plots to visualize the spread and identify potential outliers in the advertising spend and sales data.
- Why: Outlier detection is particularly important in financial datasets where extreme values can disproportionately impact business decisions.

Correlation Heatmap:

- What: To quantify and visualize relationships between variables, we plotted a heatmap.
- Why: Heatmaps simplify the interpretation of correlation matrices using color gradients. Annotations (r-values) add precision.
- We used Spearman correlation because it handles non-normal data, is robust to outliers, and captures monotonic relationships, ensuring a reliable analysis of trends between advertising spends and sales.

The near-zero correlation for Newspaper ($r = 0.05$) signaled its irrelevance, guiding its exclusion in later models.

Q-Q Plots:

- What: Q-Q (Quantile-Quantile) plots were created for each variable to assess how closely the data follows a normal distribution.
- Why: Many statistical methods assume normality. Checking for normality informs us whether transformations are needed and whether subsequent analyses (like regression) will be valid.

2.4 Normality Testing with Shapiro-Wilk

To formally test if the variables (TV, Radio, Newspaper, Sales) follow a normal distribution. Many statistical methods (e.g., Pearson correlation, linear regression) assume normality, and violating this assumption can lead to biased results.

- The Shapiro-Wilk test is a robust method for assessing normality, especially for small to moderate sample sizes.
- A significant p-value ($p<0.05$) indicates rejection of the null hypothesis (data is non-normal).

2.5 ROI Analysis

- We calculated the Return on Investment (ROI) for each advertising channel by comparing the increase in sales to the amount spent.
- ROI is a direct measure of advertising effectiveness. It helps identify which channels yield the best return, providing a clear metric for comparing different media.

2.5 Tools

Python Libraries:

- pandas for data manipulation.
- seaborn/matplotlib for publication-quality visualizations.
- statsmodels for regression and hypothesis testing.
- scipy for correlation tests.

3. Results

3.1 Descriptive statistics:

Units: For each advertising channel, the statistics are in 1000$, while for Sales, it is in Million $.

	mean	std	min	50%	max
TV	147.0425	85.8542	0.7	149.75	296.4

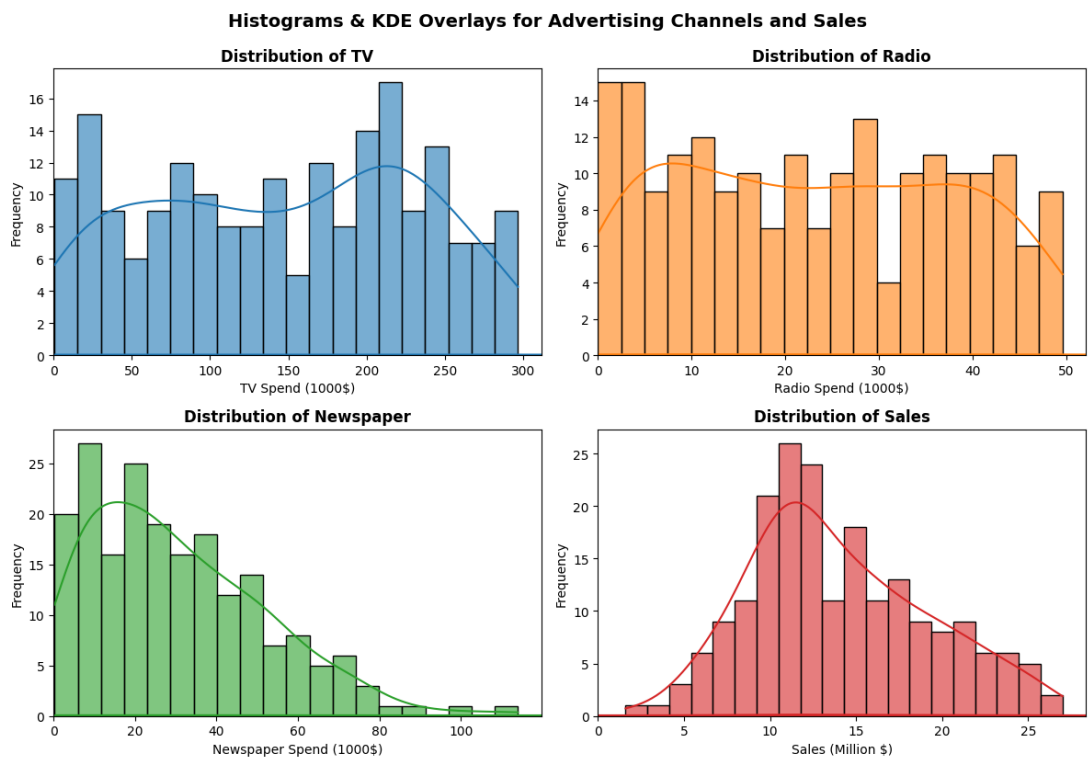
Radio	23.2640	14.8468	0	22.9	49.6
Newspaper	30.5540	21.7786	0.3	25.75	114
Sales	14.0225	5.2174	1.6	12.9	27

Some insights from the descriptive statistics about the data:

- There are 200 rows (all valid, non-null) of data for each advertisement channel and sales.
- TV has the highest mean spend (147.04), while Radio has the lowest (23.26).
- High variability shown by the TV ad expenditure.
- Sales range from 1.6 to 27.0 units, with a median of 15.05.

3.2 Visualizations

3.2.1 Histograms & Kernel Density Estimation (KDE)



Analysis

a. TV Advertising Spend

- The KDE curve reveals at least two peaks—one around 50–60 and another near 200.

- This bimodal tendency implies two distinct clusters or groups of advertisers (one set that spends moderately on TV and another that spends much more).
- The distribution does **not** appear to be heavily skewed to the right or left.

b. Radio Advertising Spend

- The highest frequency occurs in the 0.0–2.48k bin, with 21 campaigns allocating minimal Radio budgets.
- Frequency decreases gradually toward higher spends, with a secondary minor peak at 22.9–25.3k (median range).
- The KDE's smoothing effect merges neighboring bins, masking the lower peak and emphasizing central tendencies.

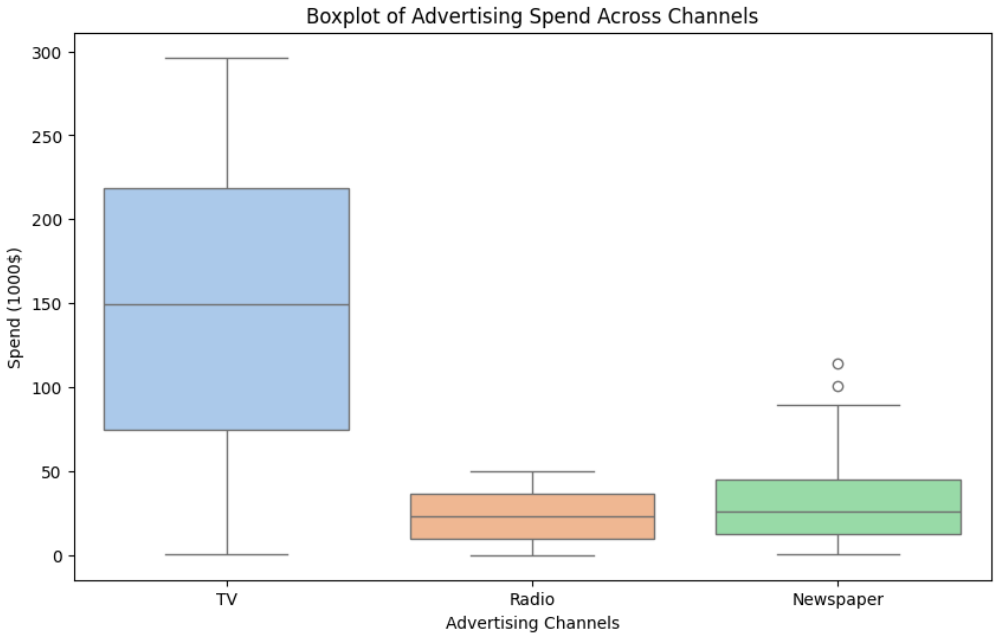
c. Newspaper Advertising Spend

- Sharp peak for KDE at 0–20k and long right tail, confirming right skewness.
- Frequency declines sharply beyond 20k, with sporadic high spends (e.g., 100–114k). This indicates the presence of outliers.

d. Sales generated

- The 11k–12k peak represents a common sales tier, possibly linked to moderate ad budgets.
- Mildly right-skewed, with clusters at 11k–12k, 15k, and 20k.
- Shows near-normal central tendency.

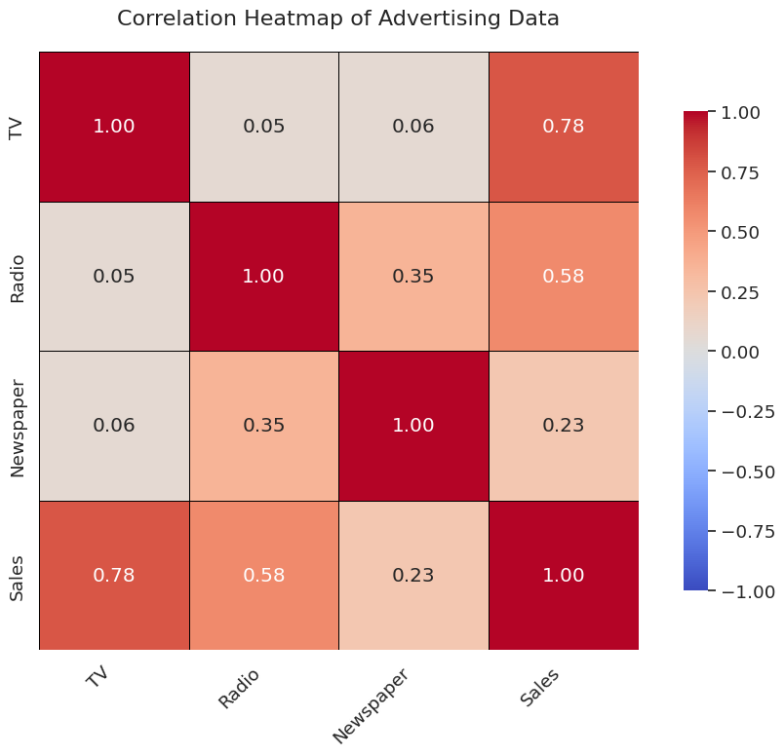
3.2.2 Box plot



Analysis:

- TV spending is consistent, with most campaigns allocating budgets between 70k and 220k.
- The wide IQR reflects strategic variability in TV investments.
- Radio budgets are tightly clustered, with most campaigns spending 10k–30k.
- Limited variability suggests standardized strategies for Radio.
- Newspaper spends 2+ campaigns with spends >70k (e.g., 100k–114k)
- Newspaper spending is highly variable, with most campaigns allocating 0k–40k.
- Outliers indicate niche strategies or data anomalies (e.g., regional campaigns).

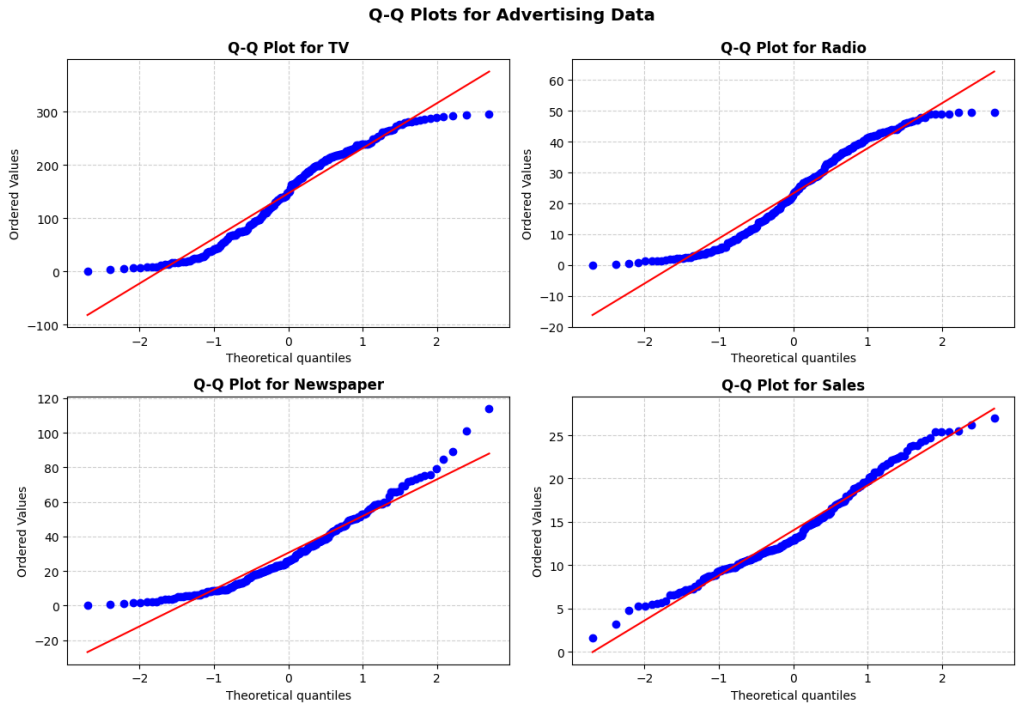
3.2.3 Correlation Heatmap



Analysis of the plot:

- TV advertising has the strongest impact on sales.
- For every dollar spent on TV ads, sales increase significantly.
- Radio advertising has a moderate impact on sales.
- It complements TV ads but is less effective on its own.
- TV spending is independent of Radio and Newspaper spending.
- Companies that spend more on Radio tend to spend slightly more on Newspaper.

3.2.4 Q-Q plots:



Analysis of the plot

- **For TV:**
  - Q-Q Plot Shape: Points deviate significantly from the theoretical line at both ends.
  - Right Tail: Points curve downwards, showing fewer low spends than expected in a normal distribution.
  - Left Tail: Points curve upward, showing skewness in the data.
- **For Radio:**
  - Points follow the theoretical line closely in the center but deviate slightly at the tails.
  - Tails: Light tails (points fall below the line), indicating fewer extreme values than expected in a normal distribution.
    - Left Tail: Points curve upward, showing right skewness in the data.
- **For Newspaper**
  - Points deviate sharply from the theoretical line at the upper tail.
  - Right Tail: Points rise steeply, indicating heavy skewness and outliers.
  - Left Tail: Points align closely with the line, showing normal behavior for lower spends.
  - Newspaper outliers may represent niche campaigns.

- **For Sales**
    - Points follow the theoretical line closely in the center but deviate slightly at the tails.
    - Left Tail: Mild upward curve, indicating light right skewness.
    - Right Tail: Points align well with the line.

3.2.5 Shapiro-Wilk test (p=0.05)

Variable	Test Statistic	-value	Conclusion
TV	0.9495	.000002	NOT normally distributed
Radio	0.9440	.000001	NOT normally distributed
Newspaper	0.9364	.000000	NOT normally distributed
Sales	0.9760	.001683	NOT normally distributed

4. **Question Specific Analysis:**

4.1 Which platform yields the highest return on investment for advertising campaigns?

Ans: To calculate the Return on Investment (ROI) for each advertising channel (TV, Radio, Newspaper), we need to determine how much sales revenue is generated per dollar spent on each channel. ROI is calculated as:

$$\text{ROI} = \frac{\text{Additional Sales from Channel} - \text{Ad Spend on Channel}}{\text{Ad Spend on Channel}} \times 100$$

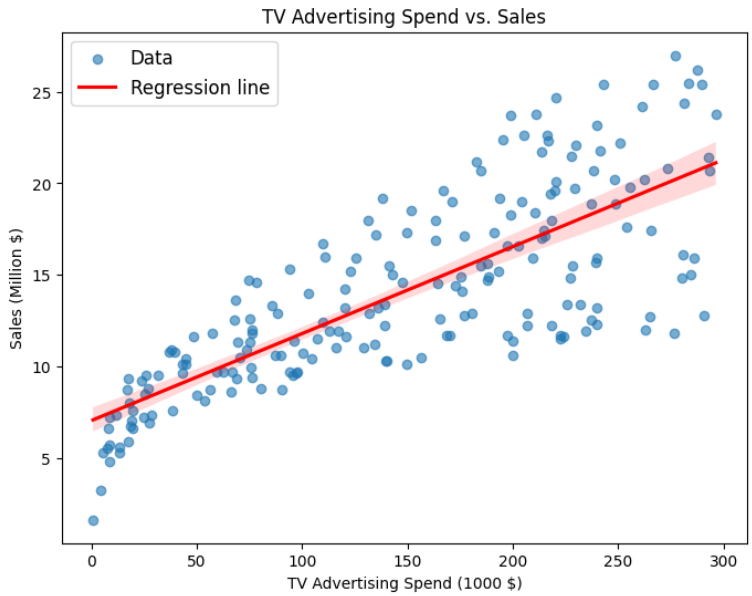
However, since we don't have explicit information about the additional sales directly attributable to each channel, we can estimate ROI using the correlation coefficients and average spending. We can simplify the formula to get: Additional Sales from channel=r×Average Sales, where r is correlation coefficient of that channel with respect to sales. ROI for each channel is as follows:

ROI for TV: 7534.95%
ROI for Radio: 33310.95%
ROI for Newspaper: 8845.78%

- Radio yields the highest ROI followed by newspapers and then TV.
- Conclusion: While TV ads drive sales the most, radio advertising provides the best cost-effective ROI.

4.2 Is there a general trend between TV advertising spend and sales?

Ans: General trend between TV advertising spend and sales can be analyzed by using a scatter plot. For reference, I have put a regression line to get a clear idea about the trend.



- As TV spends increase, Sales also increase.
- The regression line (red) shows a clear upward trend.
- At higher spend on TV advertisements, the variance increases, thus the data being more scattered around the regression line.

We can also use Pearson's correlation coefficient to get an idea on the trend:

- Correlation between TV Spend and Sales: 0.78
- Clearly, a high positive correlation coefficient indicates a positive (upward trend).

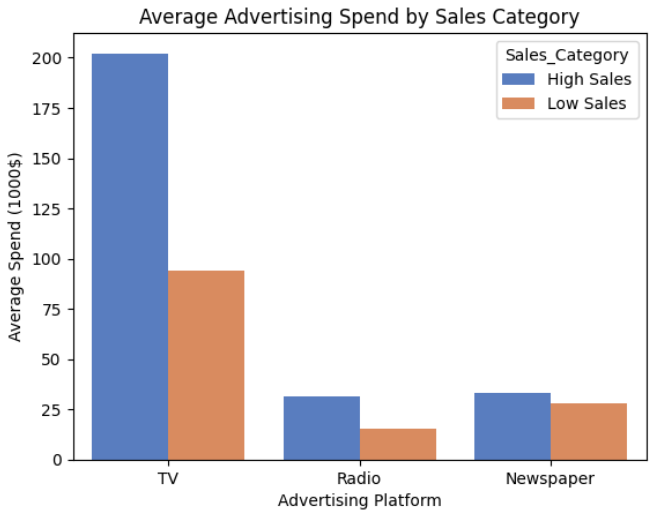
4.3 What is the relationship between ad expenditure and sales?

Ans. A general relationship between each advertising channel and sales can be found using correlation. We have already analyzed the correlation in section 3.2.3 (pls refer to the section for the answer).

For the relationship between total expenditure and sales, we can use Pearson's correlation coefficient, the correlation between total ad expenditure and Sales is 0.87. This is a high, positive correlation. (A scatter to support this is given in the jupyter notebook.)

4.4 How do the average advertising spends compare when sales are high versus when sales are low?

Ans: For this, we divide the data with respect to the median of the sales, splitting it into high and low sales group and then visualize it for each channel.



- TV and Radio spend are significantly higher in the high-sales group.
- Newspaper spends show no difference

Some additional questions:

4.5 How does variance for each channel in spending compare when sales are high vs when sales are low?

Ans: To calculate the variances, we can use the data divided into the low and high sales and calculate variance for each channel. The results are as follows:

Platform	High-sales	Low-sales	Higher Var Group
TV	3225.426721	5643.708837	Low Sales
Radio	132.031186	184.426295	Low Sales
Newspaper	508.652288	431.342059	High Sales

There is a high variance in advertising spend in case of low sales.

4.6 How does sales performance vary across different levels of TV advertisement spending?

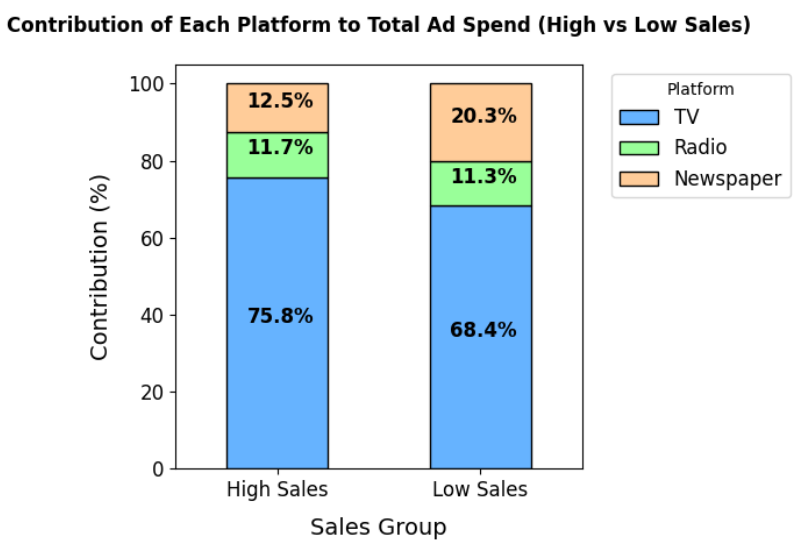
Ans: To compare the sales performance across multiple spending levels for TV advertisements, we would first divide the TV expenditure data into 3 parts (low (0-50k), mid(50-150k), and high expenditure(150k+) and then compare the mean of their corresponding sales.

TV Exp. Level	Sales avg (M $)
Low	7.705405
Medium	12.168750
High	17.581818

- There is a clear positive relationship between TV spending and sales.
- Higher TV budgets are associated with significantly higher sales.
- The marginal gain in sales is higher when moving from Medium to High TV spending compared to Low to Medium. This suggests that higher TV budgets yield disproportionately greater returns.

4.7 How does the allocation of advertising budgets across platforms (TV, Radio, Newspaper) differ between high-sales and low-sales campaigns?

Ans:



5. Conclusions and future work

Future work regarding the project would include:

- Normalizing the data for further analysis
- Apply regression to find the effect of each advertisement channel individually on sales.
- Run various inferential statistical tests to find significance of various results obtained from the data.
- Understand which combination of variables produces optimal result.
- Find if there exists diminishing return with respect to the amount spent on advertising.